

Headline part IX: the local leading-density coefficient

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Abstract

Paper-facing wrappers for the tangentially integrated normal block (unit 189). For a compact tangential set K with an integrable signed density q and a continuous amplitude $\eta(v, u)$: the mixed-ratio face-supported coefficient spelled out (`headline_tangential_normal_moment`); the equal-ratio *local leading-density* formula

$$\frac{\int_K q(v) \int_{[0,1]^{m+1}} \eta(v, u) u^h e^{-\beta N u^{2k}} du dv}{N^{-\lambda} (\log N)^m} \longrightarrow \frac{\Gamma(\lambda) \beta^{-\lambda}}{m! \prod_i 2k_i} \int_K q(v) \eta(v, 0) dv$$

(`headline_tangential_normal_moment_equal`), the chart-level precursor of the paper's $\Gamma(\lambda)/(m-1)! a_I \int_{S_I} (\phi \circ \pi) c_0 |dv|$ with c_0 and $\phi \circ \pi$ absorbed in $q\eta$; and the equivalence form under a nonzero integrated coefficient (`headline_tangential_normal_moment_equiv`). Zero sorry/axiom.

1 Paper-facing meaning

Astra #17's "leading-density corollary": at the level of one normal block in fixed coordinates, the paper's leading coefficient structure — a Gamma/Jacobian constant times an integral over the tangential variables of the amplitude restricted to the normal origin — is now a theorem when all normal ratios are equal, and is replaced by the face-supported integral when they are mixed. The identification of $\int_K q(v) \cdot dv$ with $\int_{S_I} c_0 |dv|$ (chart, density, partition of unity) is outside the formalised boundary and is not claimed.

2 Lean notes

- Pointwise goals produced by `integral_congr_ae` (`Eventually.of_forall fun v => ?_`) are beta-redexes on both sides; `beta_reduce` before `rw [amplitudeCoeff_equal ...]`.
- The mixed-ratio wrapper is a one-line `exact`: unfolding `amplitudeCoeff` applied to `fun u => (v, u)` is definitional, including the beta-redex at `faceProj u`.